

Water safety plan

Step-by-step risk management for
drinking-water suppliers

Training package



World Health
Organization



NOTE to trainer(s)

Delete this slide prior to delivering the presentation

This presentation and the associated training materials have been developed by the World Health Organization and International Water Association (2025) for adaptation and use by those conducting water safety planning training workshops.

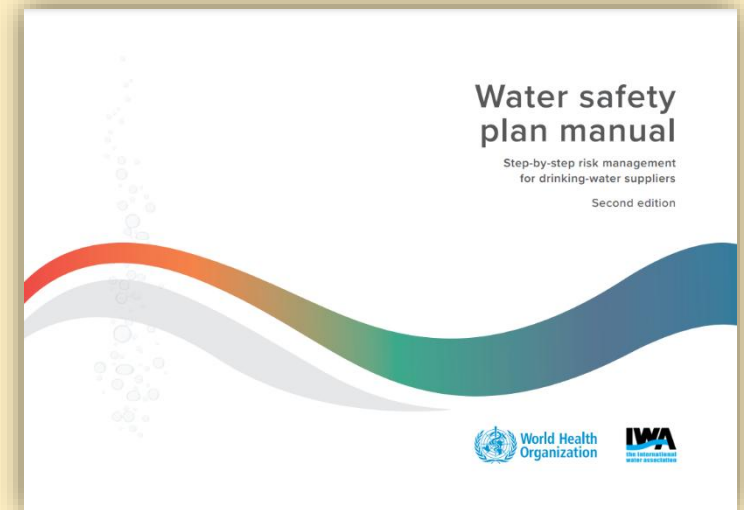
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NOTE to trainer(s)

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- 💧 These training materials are based on *Water safety plan manual*, 2nd edition (WHO & IWA, 2023).
- 💧 These materials have been developed for a **4-day** training event incorporating a partial-day field visit, but they can be adapted to accommodate other training durations and approaches, as preferred.
- 💧 Customization of these training materials to reflect the local context will contribute greatly to positive outcomes and is highly recommended.



TRAINER introductions



Insert
trainer
photo

Trainer name(s)
Affiliation

WORKSHOP objectives

By the end of this workshop, all participants should ►

1. Appreciate the value of water safety planning
2. Understand water safety plan (WSP) principles and steps
3. Understand how WSPs provide a framework to strengthen equitable access to safe drinking-water supplies that are more resilient to climate change
4. Have sufficient knowledge and confidence to begin to develop and implement WSPs

This workshop is an introduction to WSP principles and practices

WORKSHOP approach

This workshop will be ►

PRACTICAL

Practice the theory introduced **in both classroom and field**

PARTICIPATORY

Engage with activities, share ideas, ask questions

INTENSIVE

Often a 5+ day training!

WORKSHOP **overview**

DAY 1 ▶ WSP introduction, modules 2-3

DAY 2 ▶ Modules 4-7, field trip briefing

DAY 3 ▶ Field trip

DAY 4 ▶ Field trip sharing, modules 8-10 and module 1, wrap-up, next steps

RESOURCE **review**

Resources at your tables ►



**Workshop
program**



**Participant
Workbook**

PARTICIPANT introductions

Who's who in the room?

Tell us about yourself in max. 10 seconds

1. What is your name?
2. What is your organization and job title?
3. What is your WSP experience?

Answer either: new to WSPs, some experience, or a lot of experience

4. What is one fun fact about yourself?

e.g. hobby, favourite movie, favourite food, phobia, languages you speak...



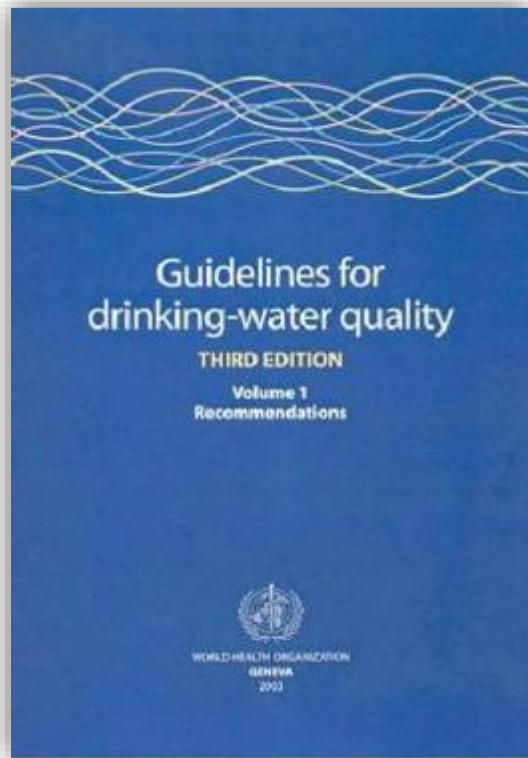
WSP introduction

- 💧 WSP background
- 💧 What is a WSP?
- 💧 Why do we need WSPs?
- 💧 WSP drivers
- 💧 Benefits of WSPs
- 💧 Global WSP uptake



WSP background

WSPs introduced by WHO in *Guidelines for drinking-water quality* (3rd Edition, 2004)



“The most effective means of consistently ensuring the safety of a drinking water supply.”

WHAT is a WSP?

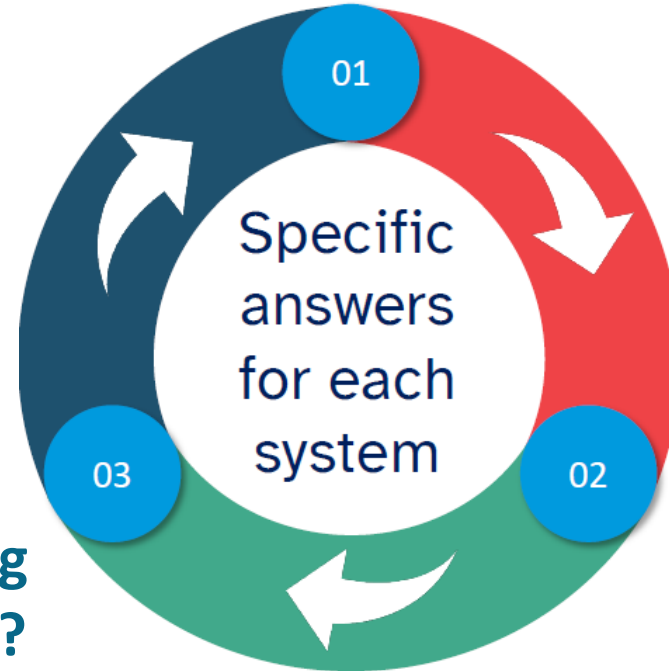
WATER SAFETY PLAN ►

A proactive **risk assessment and risk management** approach to help ensure drinking-water safety, encompassing the **entire drinking-water supply**, from catchment to consumer



Water safety planning **asks three questions**

What are the priority risks?



How can we confirm the ongoing effectiveness of the WSP?

How can we better manage the risks?

WSPs involve...



Preventing
contamination at
the source



Removing/
reducing
contamination by
water treatment



Preventing
recontamination
in storage and
distribution



Preventing
recontamination at
the household

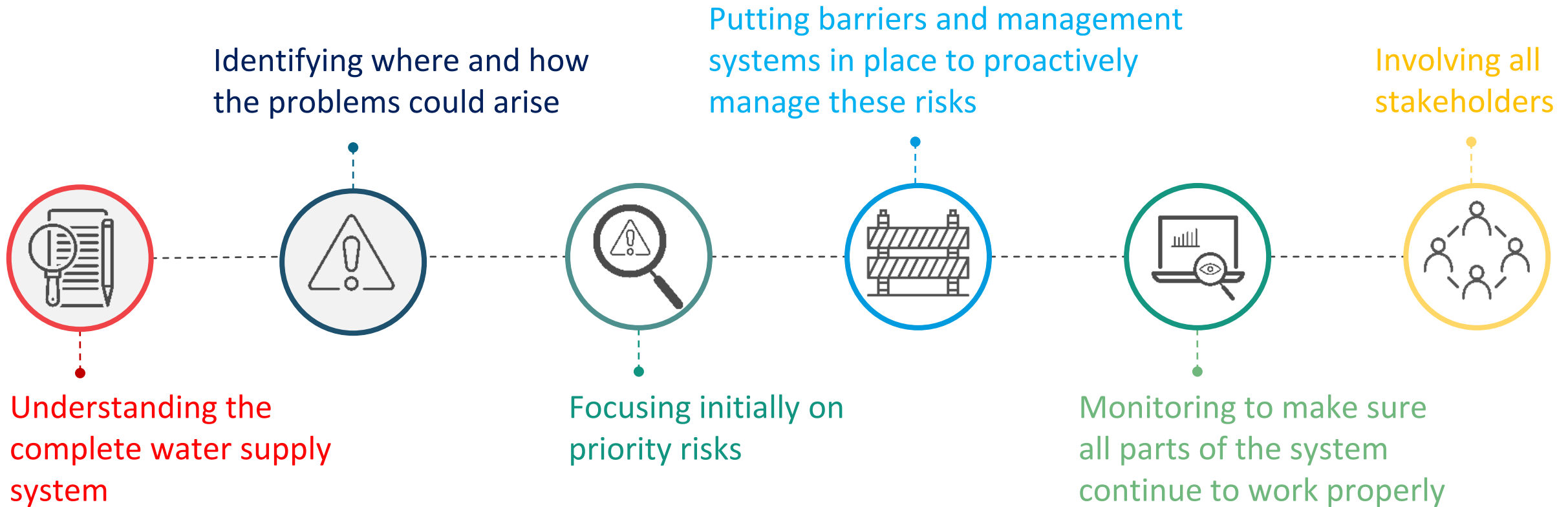
Picture 1: Olivier Girard/CIFOR via flickr.com (CC BY-NC-ND 2.0)

Pictures 2-4: WHO/Rory Moses McKeown

WHAT is a WSP?

A proactive management system to ensure safe drinking-water by ►

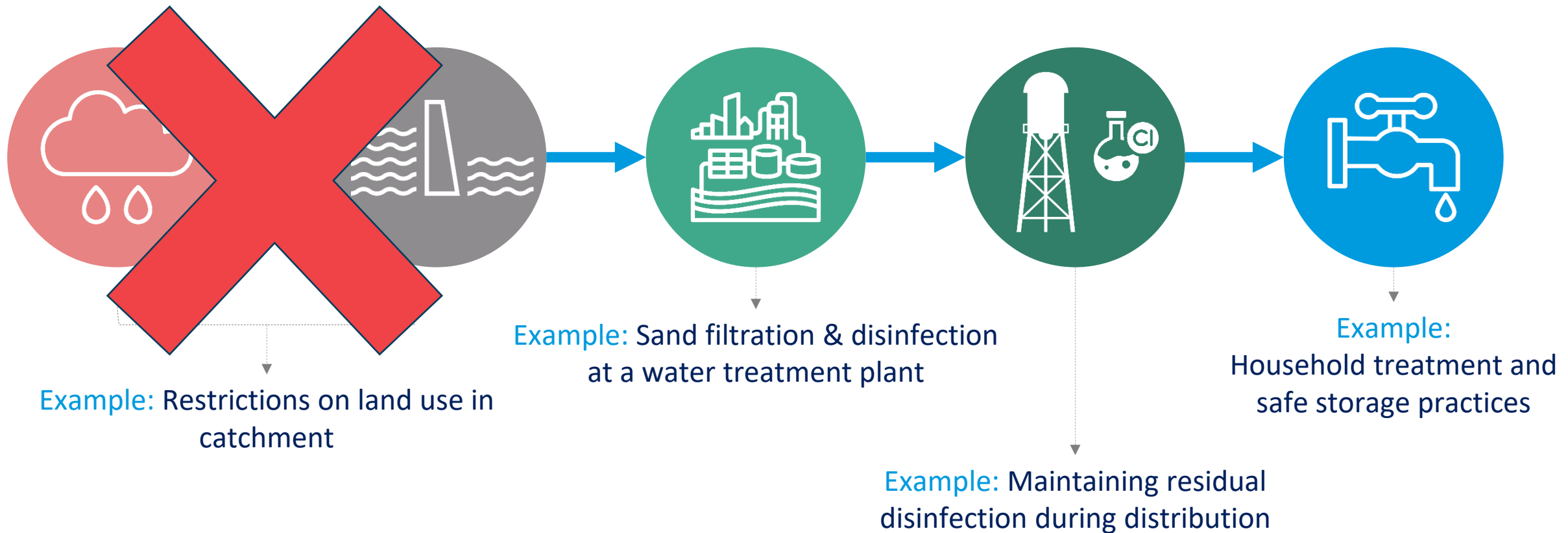
5 ►



WSPs use a “multi-barrier approach”

Multiple-barriers in action to prevent microbial contamination ►

1 ►



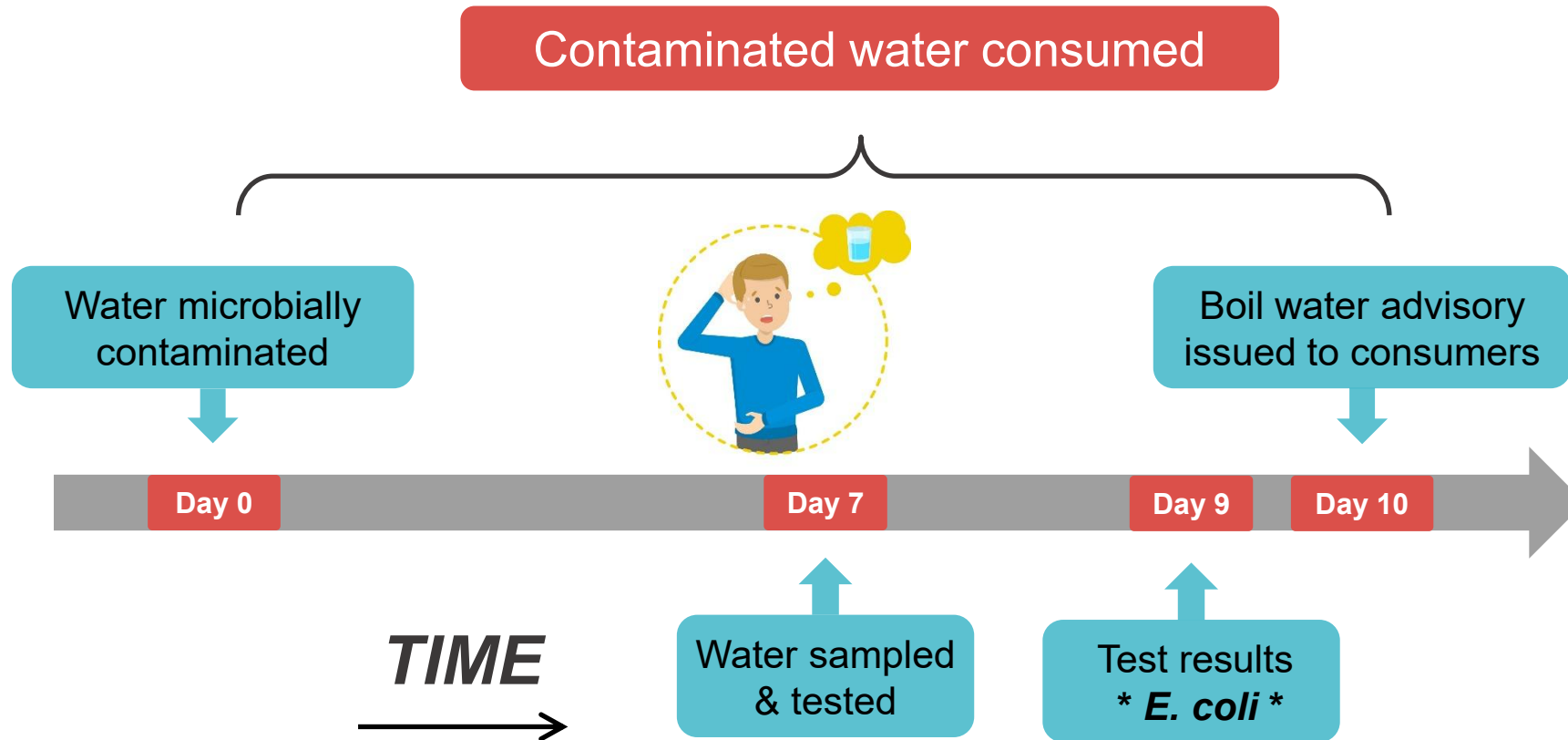
WHY do we need WSPs?

Without a risk management approach, water suppliers must rely only on
END-PRODUCT TESTING
to confirm water safety



WHY do we need WSPs?

End-product testing scenario



EXERCISE:

Limitations of end-product testing



10 minutes

(5 to complete; 5 to discuss)

Working in groups, make a list of the problems with relying on end-product testing to ensure water safety.



After exercise, review examples on following slide...

WHY do we need WSPs?

Limitations of end-product testing include ►

- 💧 **Reactive approach**
(problem has already occurred)
- 💧 **Test results provide a “spot check” only**
(problems can be missed in space and time)
- 💧 **Limited laboratory/equipment capacity**
- 💧 **Testing can be very expensive**
- 💧 **May not be clear what went wrong, where and when**
(supplier may not know how to correct problem)

**TOO LITTLE,
TOO LATE!**

WHY **do we need WSPs?**

Water quality testing is an important component of water supply system management...

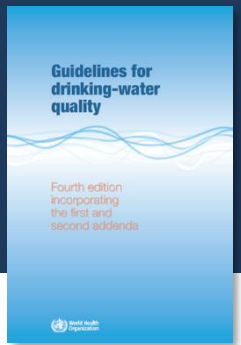
...but it **is not enough to ensure water safety.**



Need a **proactive** approach to ensure water safety through good management of the complete water supply system → **WSP**

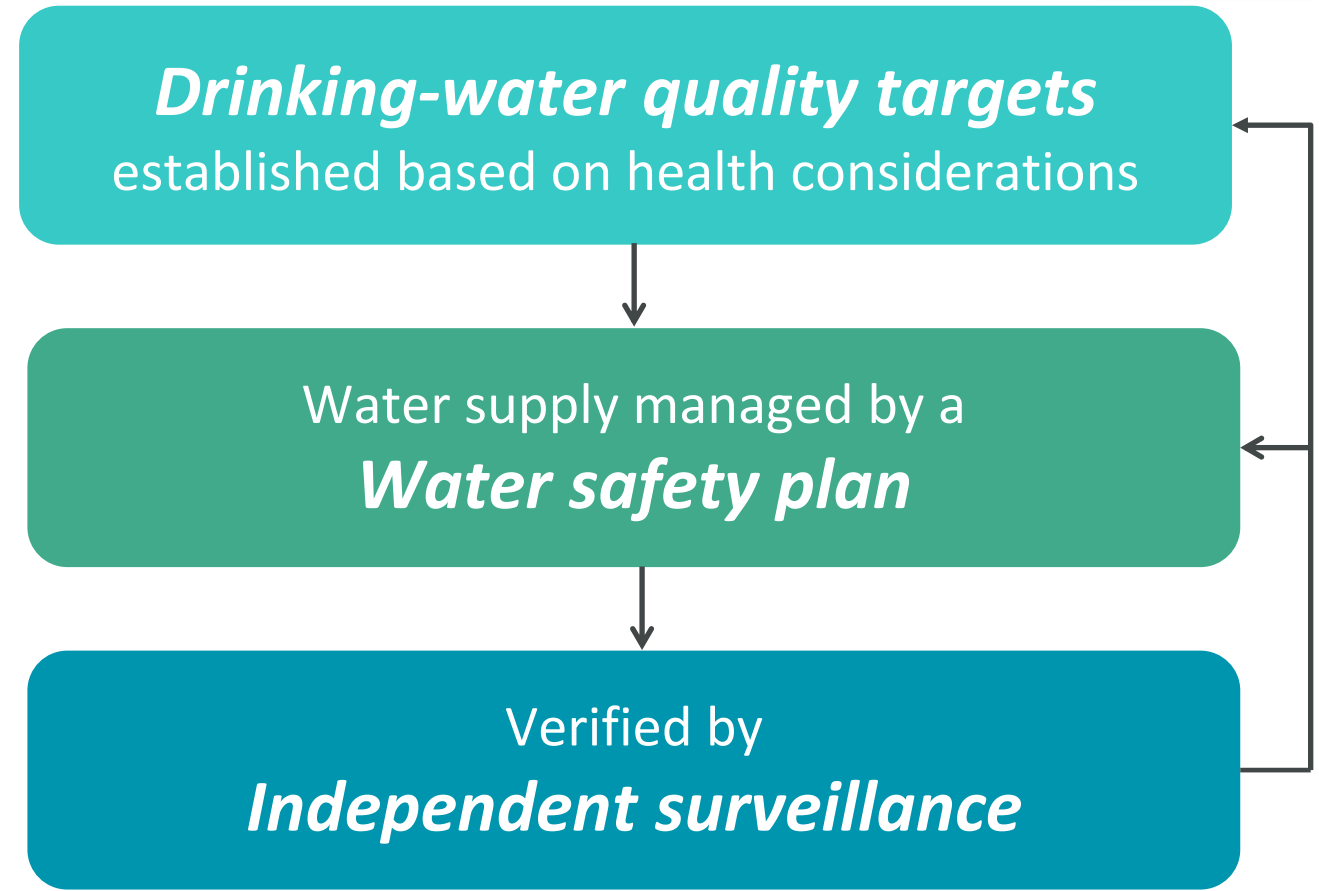
WSPs play a key role in delivering safe drinking-water

Framework for safe drinking-water ►



WSPs are one of three functions recommended by WHO to ensure drinking-water safety

⇒ can help water suppliers to achieve drinking-water quality targets



After *Guidelines for drinking-water quality* (WHO, 2022)

WSPs and the Sustainable Development Goals (SDGs)



Increased focus on
**proactive
management**
to ensure safe
delivery of drinking-
water services

WSP drivers

Regional or national drivers, such as drinking-water quality policies and legal instruments, for example:

- 💧 National standards requiring WSPs in Lao People's Democratic Republic
- 💧 European Union Drinking Water Directive
- 💧 Sri Lanka national policy on drinking-water safety

WSP policies or regulations have been formally approved in over 60 countries*

*WHO global tracking database (as at 2022)



BENEFITS of WSPs

WSP OUTCOMES

- 💧 Improved system understanding
- 💧 Improved stakeholder collaboration
- 💧 Prioritized improvement needs
- 💧 Improved operations and management
- 💧 Cost savings via improved operational efficiencies
- 💧 Reductions in non-revenue water
- 💧 Improved infrastructure
- 💧 Leveraged financial support

*Leading
to*

WSP IMPACTS

- 💧 Improved water quality
- 💧 Improved service delivery
- 💧 Improved health

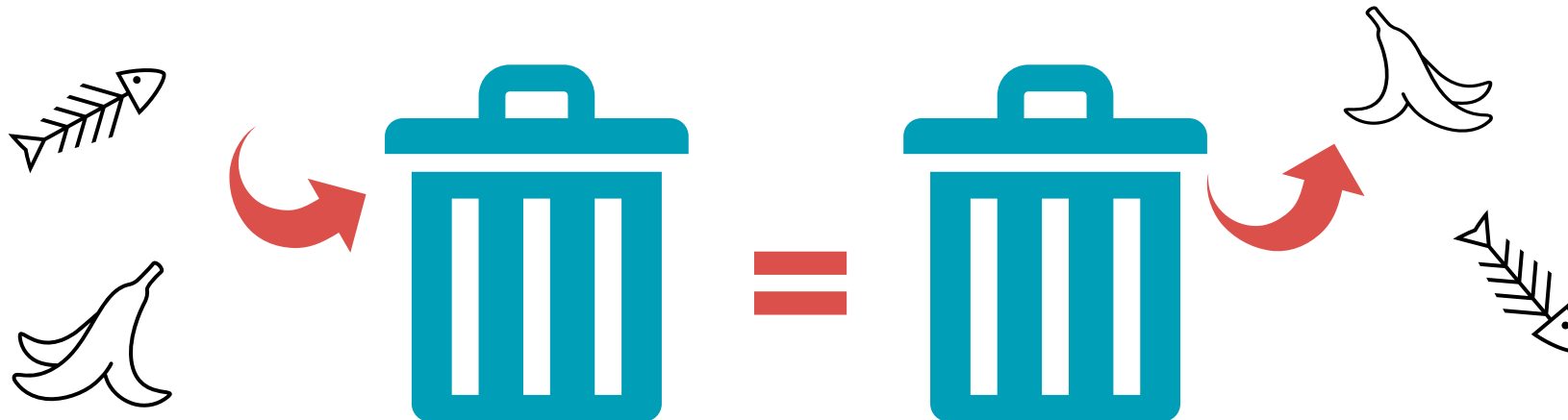
✓ Suppliers ✓ Consumers ✓ Health authorities

MANY WSP benefits, however GIGO principle applies...

Garbage
In
Garbage
Out

We get out of WSPs what we put into them.

Attitude is key!



WSPs provide resilience to future shocks and uncertainty

WSPs support future planning for foreseen and unforeseen events

WSPs can strengthen the resilience of the water supply e.g.

- ☑ Support business continuity planning
- ☑ Strengthen emergency preparedness and response planning
- ☑ Help ensure robust supply chains (e.g. treatment additives, consumable reagents for testing)



WSPs and building climate resilience

WSPs can help to prioritize and manage current and future climate threats

- Considers the implications of climate change at each stage of the water supply



Example climate impacts	Hazardous event	Example improvement measures needed
Increased temperature	Reduced water quantity due to reduced rainfall and increased demand	Catchment/source Provision of additional boreholes to supplement existing water source
Reduced precipitation		Treatment Backwash water recovery program to minimize water wastage
Increased drought		Distribution/storage Leak detection Mains repair/replacement programme Household User education on water conservation during drought periods

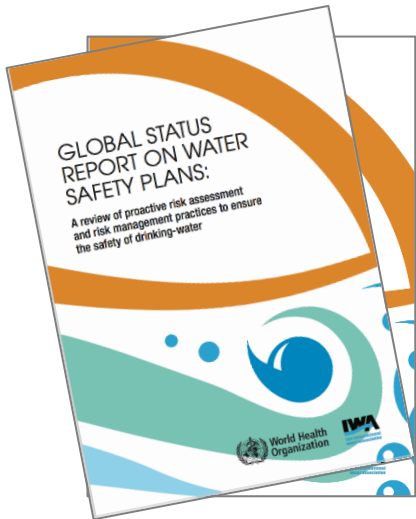
WSPs and enhancing equitable access to safe drinking-water

Vulnerable and disadvantaged groups can be explicitly considered in WSPs to help reduce inequalities e.g.

- 💧 Considering users in informal settlements when assessing risks
- 💧 Including all users in monitoring programmes
- 💧 Developing emergency response plans that consider the needs of different groups

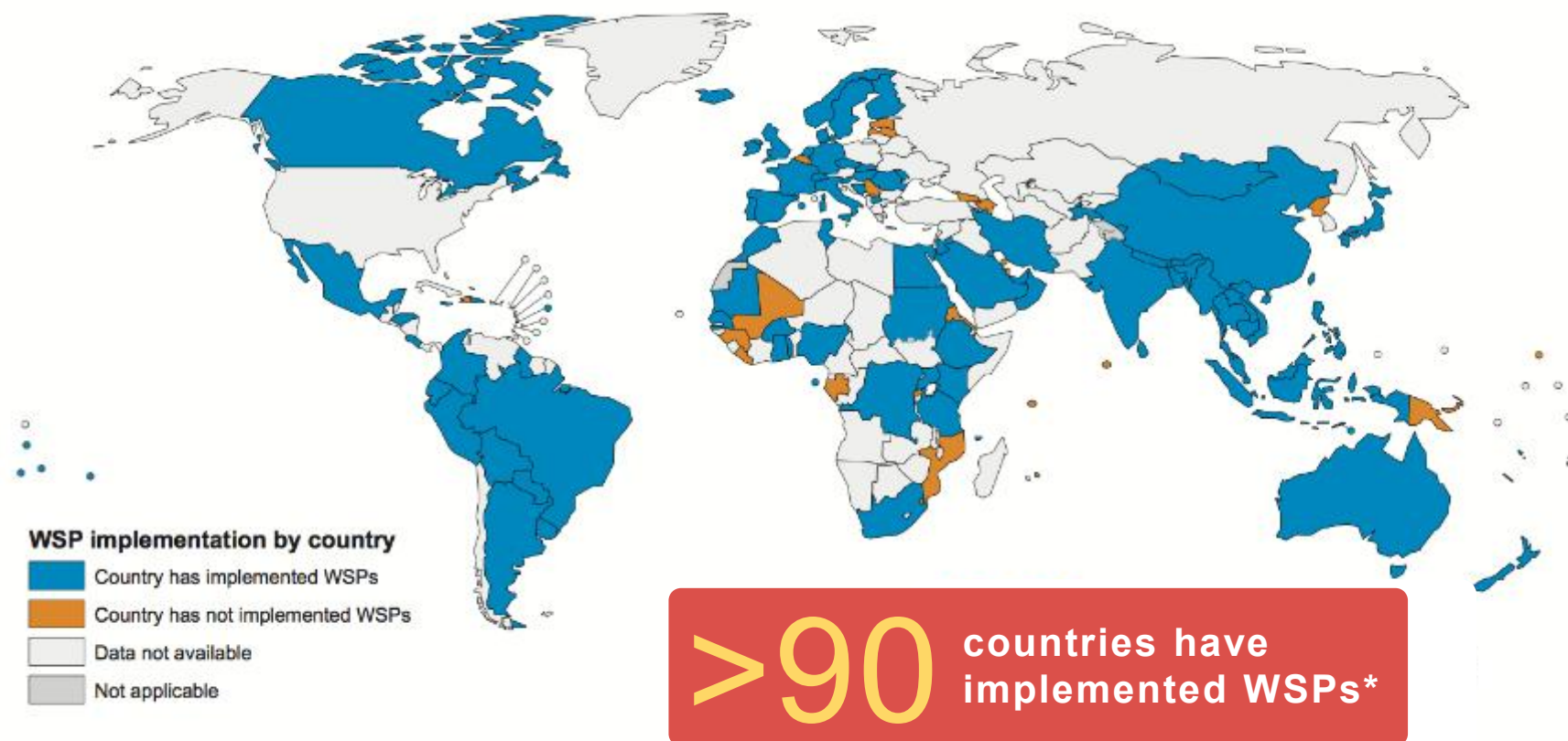


GLOBAL WSP uptake



**How many countries
have implemented WSPs?**

GLOBAL WSP uptake



** Or equivalent risk assessment and risk management approaches.*

WHO/IWA (2017) Global status report on water safety plans. Geneva, Switzerland.

ACTIVITY 1:

Water quality and health

Activity 1 – Workbook page 3



15 minutes

(10 to complete; 5 to review)

Working in groups, and as a team, read the instructions in the **Participant Workbook** (page 3).

Choosing from the list provided, complete the table by matching each water-related hazard with the signs and source of the hazard. Review answers together.

Example on next slide...



EXERCISE:

Water quality and health

1 ➡

EXAMPLE:

Potential signs	Possible hazard	Possible contamination source (or hazardous event)
Acute health issues		
Diarrhoea and dysentery (including occasional outbreaks of cholera and typhoid fever) and other waterborne infections such as hepatitis are widespread within the community, particularly affecting the young, old and health compromised	Microbial pathogens	Open defecation or nearby sanitation facilities; agriculture (use of manure) or wildlife; dirty water with suspended particles such as silt, clay or organic matter (often from flood waters or following rainstorms)

After exercise, review answers on following slides...

Potential signs	Possible hazard	Possible contamination source (or hazardous event)
Acute health issues		
Diarrhoea and dysentery (including occasional outbreaks of cholera and typhoid fever) and other waterborne infections such as hepatitis are widespread within the community, particularly affecting the young, old and health compromised	Microbial pathogens	Open defecation or nearby sanitation facilities; agriculture (use of manure) or wildlife; dirty water with suspended particles such as silt, clay or organic matter (often from flood waters or following rainstorms)
Methaemoglobinaemia in bottle-fed infants (or "blue baby syndrome")		Sewage discharges; poorly maintained septic tank; animal manure and runoff from agriculture

Potential signs	Possible hazard	Possible contamination source (or hazardous event)
Chronic health issues		
Mottling and staining of teeth in young children and teenagers; brittle bones and crippling		Naturally occurring in some groundwaters
Pigmentation changes (melanosis) and thickening of the skin (hyperkeratosis), increased rates of cancers		Naturally occurring in some groundwaters
Skin irritation (skin rash, hives, itchy eyes and throat); tingling around the mouth and fingertips; slurred speech; animals who drink the water may die		High nutrient levels in warm and stagnant surface water (ponds, tanks)

Potential signs	Possible hazard	Possible contamination source (or hazardous event)
Acceptability issues		
Stains on fixtures or laundry; coloured water with metallic taste		Naturally occurring in some groundwaters; may result from corroding pipes in the distribution system or from "overturning" of reservoirs
Unpleasantly salty taste (may pose health concern to those on sodium-restricted diets)		Naturally occurring in some groundwaters; may be from seawater (coastal areas) or caused by evaporation residue in irrigated areas (hot climates)
Rotten egg odour and taste; corrosive black spots in pipes		Naturally occurring in some groundwaters; could indicate industrial waste, oil, coal or stagnant water

Potential signs	Possible hazard	Possible contamination source (or hazardous event)
Acceptability issues		
Brown-coloured water without particles (not direct harmful to health but could result in high levels of disinfection by-products if water is chlorinated)		Naturally occurring in some surface waters from lakes or rivers with submerged vegetation
Soap does not lather; white scale builds up on pots or kettles when water is heated (not harmful to health, but may make water difficult to treat and use)		Usually from limestone or chalk aquifers

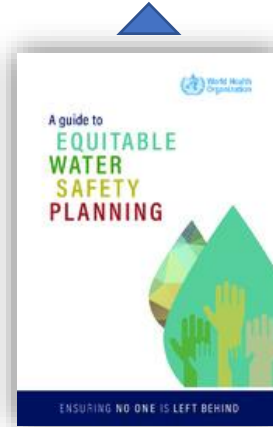
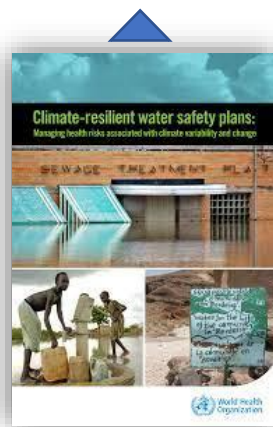
BASIS of this WSP training

WHO/IWA (2023)



Reflects over 15 years of practical experience since the first edition (2009)

Streamlined integration of enhancing climate resilience and equity in WSPs



10 WSP modules

DAY 1	Mod 2: Describing the system
	Mod 3: Identifying hazards and hazardous events
DAY 2	Mod 4: Validating existing control measures and assessing risks
	Mod 5: Planning for improvement
	Mod 6: Monitoring control measures
	Mod 7: Verifying the effectiveness of water safety planning
DAY 4	Mod 8: Strengthening management procedures
	Mod 9: Strengthening WSP supporting programmes
	Mod 10: Reviewing and updating the WSP
	Mod 1: Assembling the WSP team

MODULE

Assembling the WSP team



MODULE 1

Will discuss on **Day 4**



Outlines roles and responsibilities for the team, and stakeholder engagement.

Participants first need to understand the full WSP process.

Only then can you fully appreciate who needs to be involved, when, and why, to ensure the overall undertaking is effective.

MODULE

2

Describing the system

*How does the system deliver drinking-
water from catchment to consumer?*



MODULE 2 **overview**

1

WHAT WE DO:

Develop an accurate, up-to-date description of the complete water supply system

2

WHY WE DO IT:

To contribute to system understanding and to facilitate identification of system vulnerabilities in subsequent steps

3

HOW WE DO IT:

1. Gather, document and update system information (in words and supporting diagrams)
2. Confirm the accuracy of the system description

KEY ACTION 1:

Gather, document and update system information



Describe the complete water supply system



List water quality targets / standards



Include known / suspected problems or vulnerabilities



What should we describe at each step?

INCLUDE summarized information to help identify vulnerabilities

3 ➡



Water source(s), yields, catchment land use, source water quality/issues, abstraction details, climate information/projections, etc.

Treatment processes, chemicals used, treatment capacity (versus production), treated water quality/issues, etc.

Storage tanks (number, capacity, age, material), pipeline information (age, material), tapstands, delivery tankers, kiosks, etc.

Consumer practices (collection, storage, treatment, handling), including vulnerable and disadvantaged groups, etc.

INCLUDE information on the “use” and “users” of the water



EXAMPLE:

Intended use	Intended users
The water supplied is intended for consumption, food preparation, bathing and laundry.	GENERAL POPULATION. The intended users <i>do not</i> include those that are significantly immuno-compromised or industries with special water quality needs. These groups are advised to provide additional point-of-use treatment.

What is meant by “general population”?

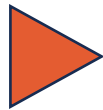
Do we need to be more specific to ensure equitable benefit?

WHY it is important to consider all users

1 ➡

If the diverse water experiences of different user groups are not considered through the WSP process, the **WSP may overlook important risks affecting some users**

E
X
A
M
P
L
E



Household tap
(most residents)



Public tap
(informal settlements)



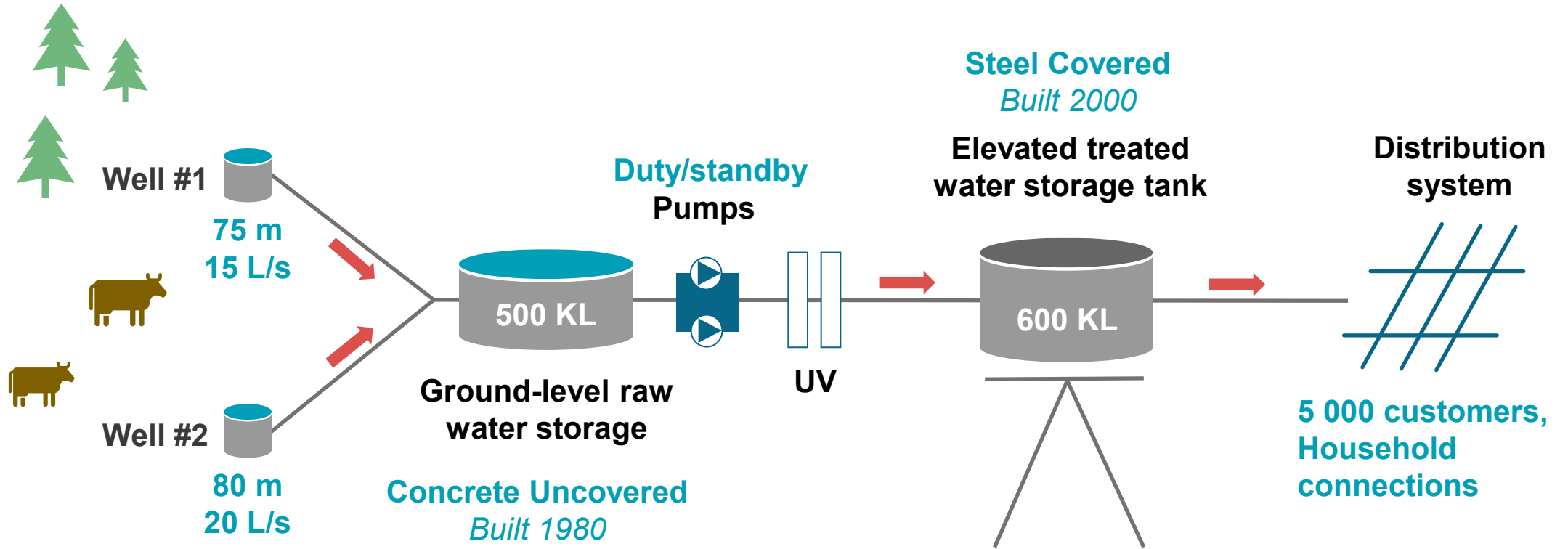
Credit: WHO/RM McKeown

Remember to consider all users, including vulnerable & disadvantaged groups

DEVELOP a system diagram

A great deal of information can be presented on a diagram

1 ➡

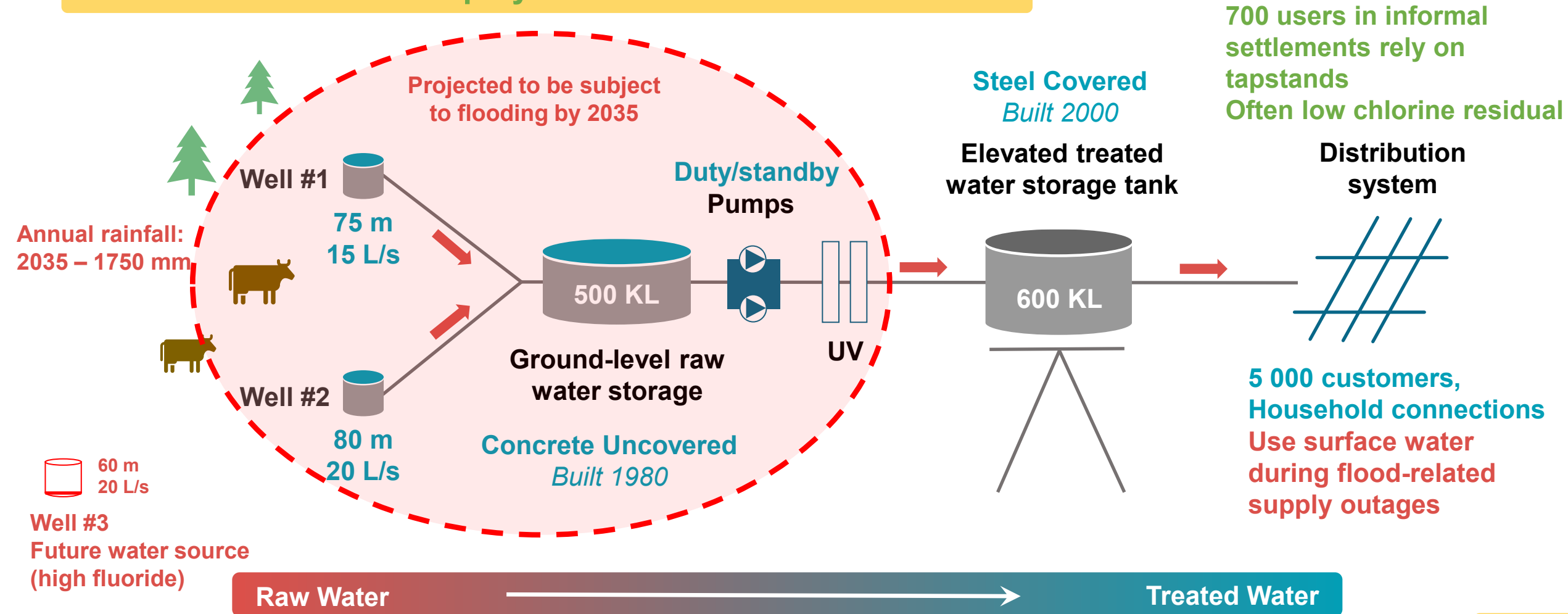


Raw Water

Treated Water

DEVELOP a system diagram

Consider climate and equity information



SYSTEM DIAGRAMS are an important part of the system description

System diagrams help to:

- 💧 strengthen conceptual understanding of the complete system
- 💧 clarify water flow, linkages, responsibilities and vulnerabilities



Create system diagrams with sufficient detail to help identify problems or potential problems

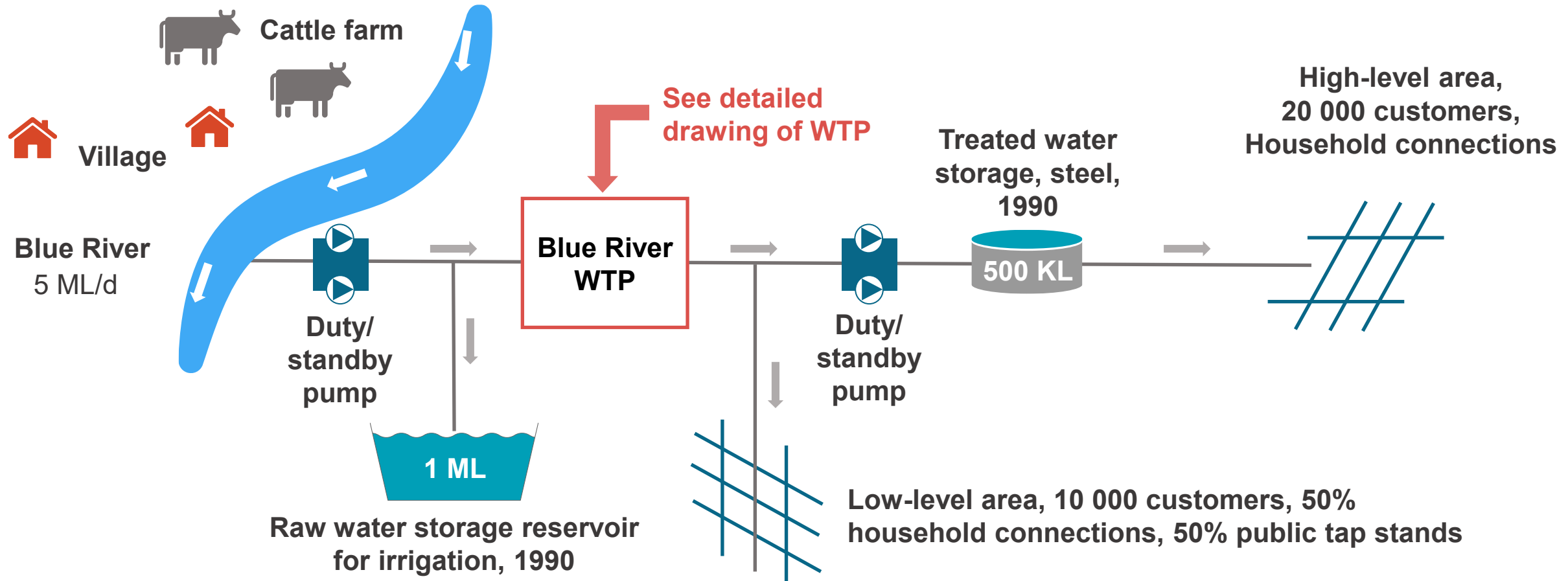
If needed, consider creating diagrams at two levels:

Level 1: system overview
(catchment to consumer)

Level 2: water treatment
plant details

LEVEL 1:

System overview



Raw Water

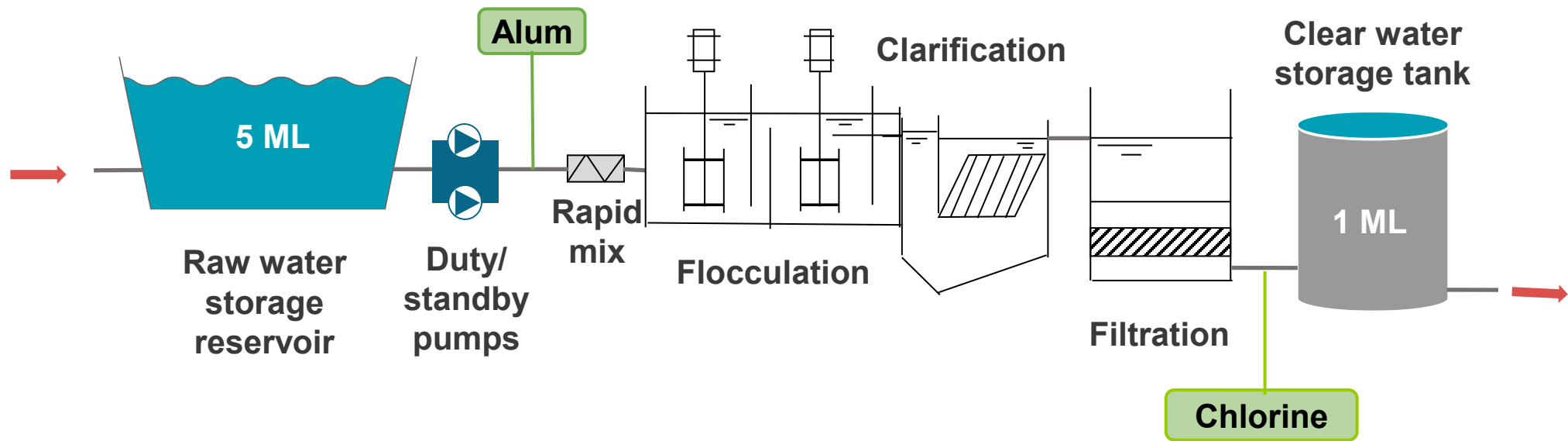
Treated Water

LEVEL 2:

Water treatment plant details

Blue River Water Treatment Plant (WTP)

(5 ML/day, built 2014)



Raw Water

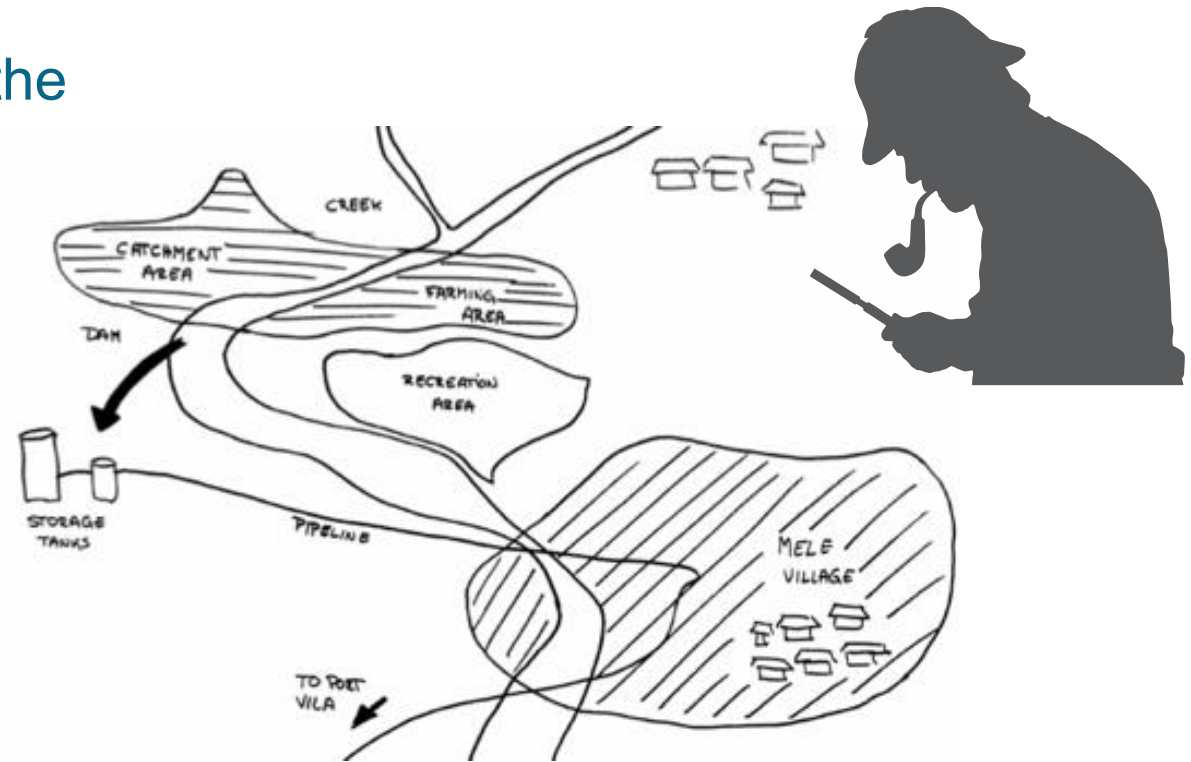
Treated Water

KEY ACTION 2:

Confirm the accuracy of the system description

Follow the flow of water and **field verify** that the description is thorough and accurate.

- ✓ Interview relevant field staff, operators or stakeholders to confirm accuracy of information
- ✓ Ensure that the system description is consistent with accurate data and reports
- ✓ Update the system description based on the outcomes of these activities



Walk the system!

Don't rely on a desktop review only

EXERCISE:

Where does the WSP end?



10 minutes

(5 in groups; 5 in plenary)



Consider this water supply scenario:

- ▶ Intermittent supply is standard operation
- ▶ Consumers generally receiving water once every one or two weeks
- ▶ Household **rooftop tanks** are part of the supply chain



Credit: WHO/RM McKeown

EXERCISE:

Where does the WSP end?



10 minutes

(5 in groups; 5 in plenary)



Working in groups, consider where the WSP should end:

**At the
water meter?**

(before rooftop storage)



At the tap?

(after rooftop storage)



**At the point
of consumption?**

(after household treatment)



ACTIVITY 2:

System diagram

Activity 2 – Workbook page 5

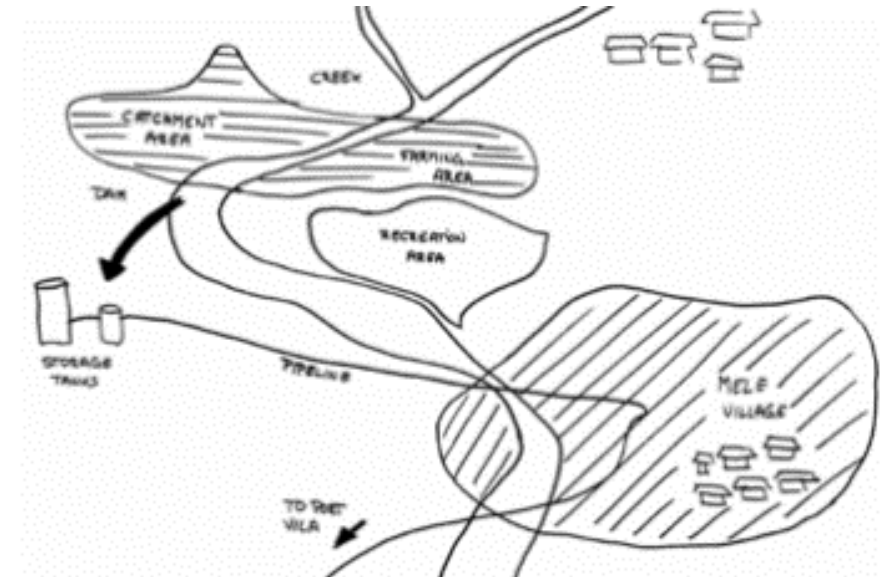


50 minutes

(40 to complete; 10 to review)

Part I: Working in groups, create system diagrams as described in the **Participant Workbook** (page 5).

Part II: Review diagrams in table group pairs assigned by the trainer.



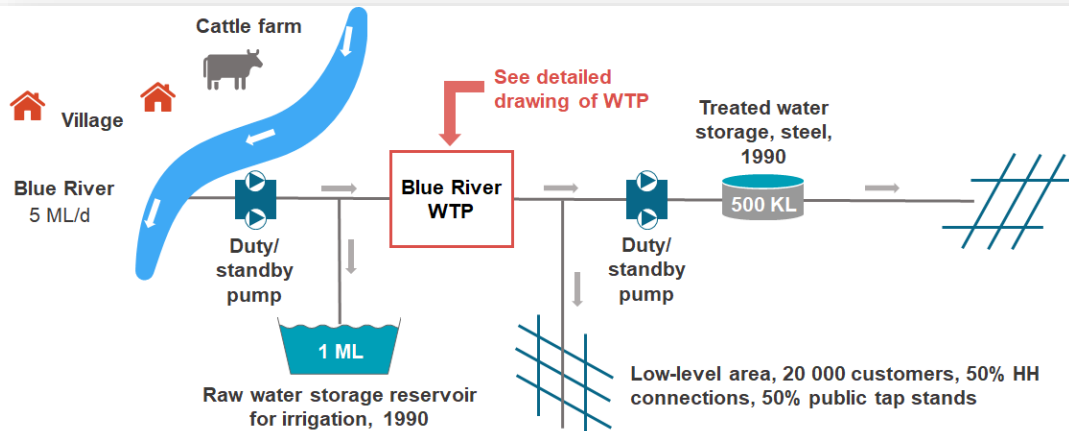
ACTIVITY 2:

System diagram – hints & tips



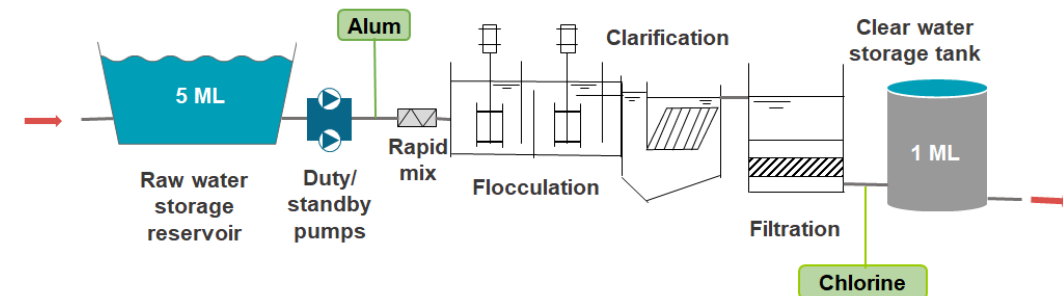
Reminder: system diagrams should include sufficient detail to help identify problems or potential problems

Level 1: system overview
(catchment to consumer)



Level 2: water treatment
plant details

Blue River Water Treatment Plant (WTP)
(5 ML/day, built 2008)



MODULE

3

Identifying hazards and hazardous events

What could go wrong?



MODULE 3 **overview**

1

WHAT WE DO:

List the issues that may threaten the safety and adequacy of the water supply

2

WHY WE DO IT:

To understand what could go wrong in the entire system to ensure important risks can be identified and managed

3

HOW WE DO IT:

For each step in the water supply chain, identify hazards and hazardous events that could contaminate or compromise the water supply

DEFINITIONS

HAZARD

A contaminant or condition that may adversely affect the supply of safe drinking-water

HAZARDOUS EVENT

An event that results in a hazard being introduced to, or inadequately removed from, the water supply



HAZARD vs. hazardous event

2 ➡

HAZARDOUS EVENT ►

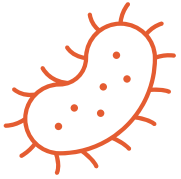
The houses are damaged because of heavy rains that destabilize the hillside, releasing the boulder



HAZARD ►
Large boulder

HAZARD types

Generally classify hazard types as ►



Microbial

e.g. pathogens such as E. coli, Legionella, Cryptosporidium, norovirus, enterovirus



Chemical

e.g. arsenic, fluoride, lead, manganese, nitrate, iron, synthetic chemicals, pesticides



Radiological

e.g. radionuclides, radium, uranium, tritium



Acceptability

e.g. taste/odour, colour, turbidity, hardness

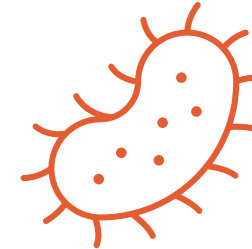
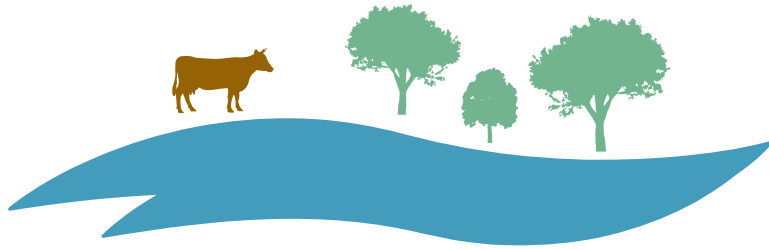


Quantity-related

e.g. insufficient water for hydration, hygiene, domestic needs

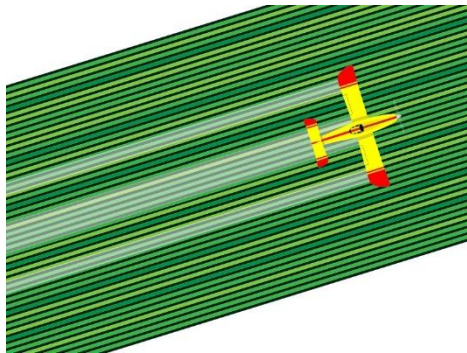
HAZARDS and hazardous events

2 ➡



HAZARDOUS EVENT ► Source water is faecally contaminated because of faecal run-off into the source water from cattle grazing

HAZARD TYPE ► Microbial



HAZARDOUS EVENT ► Source water is chemically contaminated because of pesticide run-off into source water

HAZARD TYPE ► Chemical

EXAMPLES of possible problems

3 ➡

Catchment/ source	Animal waste in runoff; industrial discharges; onsite-sanitation overflow during flood; agricultural chemicals in runoff; natural chemicals in groundwater; sediment from bushfires
Treatment	
Storage & distribution	
Consumer practices	



What are possible
problems at each
step?

DESCRIBING hazardous events

When describing hazardous events, **BE SPECIFIC!**

Clearly indicate **what hazard** could be introduced, and **how**.



TIP ►

A good hazardous event reads like this:

X happens *(to the water supply)*
because of Y

DESCRIBING hazardous events

X happens (to the water supply) because of Y

EXAMPLES:

- 💧 Source water becomes faecally contaminated (X)
because of discharge of untreated domestic waste from households (Y)
- 💧 Water in the pipe network becomes contaminated (X)
because of unsanitary pipeline repair practices (Y)
- 💧 Water is over- or under-dosed with chlorine (X)
because of insufficient operator training (Y)

X = What can happen
to the water supply

Y = How it can
happen (i.e. cause)

DESCRIBING hazardous events

2 ➡

✗ WEAK EXAMPLE ▶

Hazardous event: Agriculture in the catchment

X = What can happen to the water supply
Y = How it can happen (i.e. cause) } **Are these clear enough?**

✓ STRONGER EXAMPLES ▶

Hazardous event: The source water is faecally contaminated (X)
due to use of animal waste on crops and runoff during heavy rains (Y)

Hazardous event: The source water is chemically contaminated (X)
due to use of pesticides on crops and runoff during rains (Y)

DESCRIBING hazardous events

✗ WEAK EXAMPLE ▶

Hazardous event: Old pipes

X = What can happen to the water supply
Y = How it can happen (i.e. cause) } Not clear enough

✓ STRONGER EXAMPLES ▶

Hazardous event: Acceptability and microbial contaminants enter the pipe network (X) due to leaks in old pipes (Y)

Hazardous event: The water supply is chemically contaminated (X) due to leaching of lead from old pipes (Y)

IDENTIFYING hazardous events

When identifying hazardous events, consider each step in the water supply chain and ask:

What could potentially go wrong at this location?

- *What is wrong now, what has gone wrong in the past, and what could possibly go wrong in the future?*
- *Consider climate information & projections!*

Identifying hazardous events should involve:

- ✓ Field visits/site inspections (incl. discussions with field staff, stakeholders)
- ✓ Desktop review of Module 2 system diagrams (detailed and accurate!)
- ✓ Consideration of all user groups



EXERCISE:

Hazard brainstorm



15 minutes

Working all together in plenary, make a bullet list of some potential problems in the pictures on the next slide.

Then, translate these into clearly-defined hazardous events/hazard types.

EXAMPLE



Hazardous event ► The source water is faecally contaminated (X) due to the animals defecating in the river at the abstraction point (Y)

Hazard ► Microbial (M)

EXERCISE:

Dug well hazard brainstorm

Hazardous event: Household well water microbiologically contaminated during collection because of flooding Hazard: Microbial	Hazardous event: Ingress of contaminated surface runoff because of broken spring box Hazard: Microbial	Hazardous event: Microbial contamination from insects or rodents accessing the reservoir because of damaged screens Hazard: Microbial
Hazardous event: Water losses because of pipe leaks Hazard: Quantity	Hazardous event: Chemical contamination of water because of use of a chemical container to store household water Hazard: Chemical Acceptability	Hazardous event: Contamination of stored water because of missing tank cover Hazard: Microbial

ACTIVITY 3:

Hazard identification

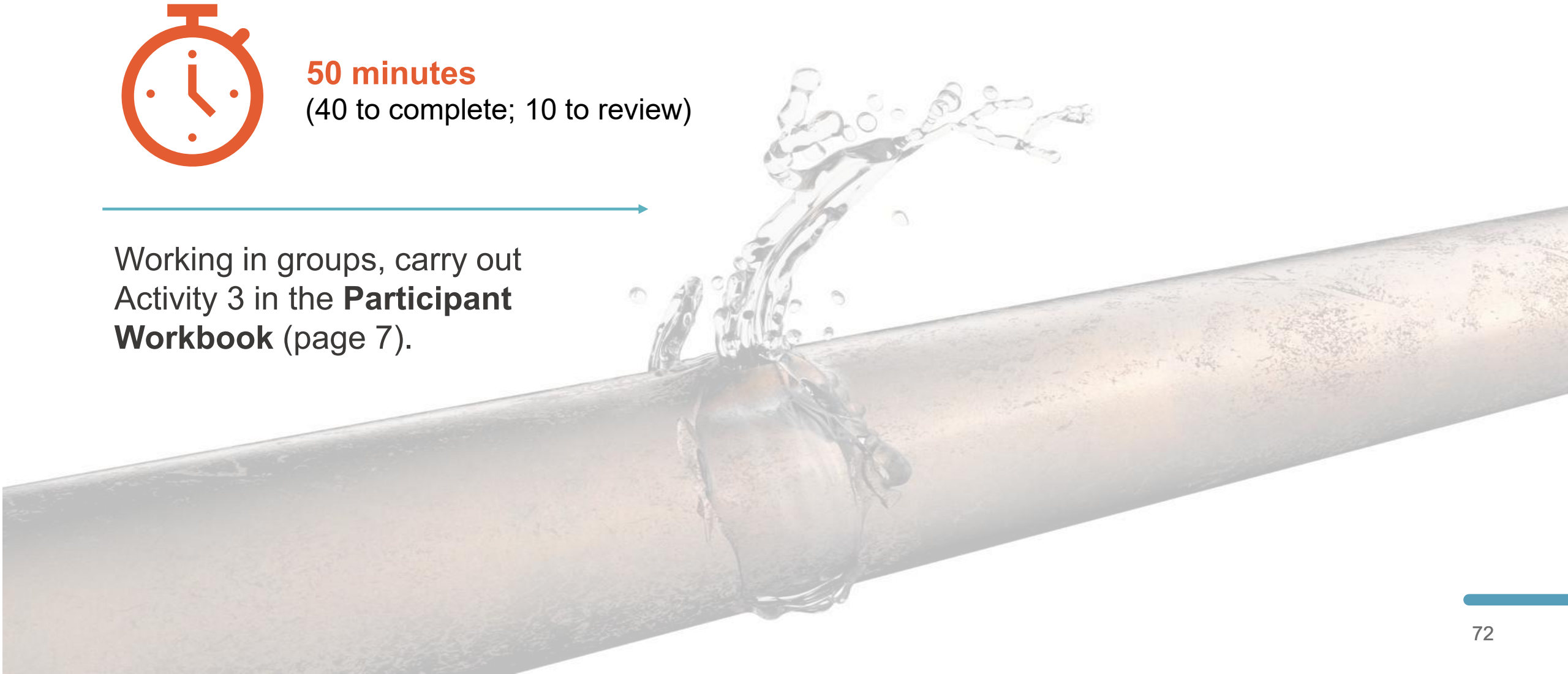
Workbook Activity 3 (page 7)



50 minutes

(40 to complete; 10 to review)

Working in groups, carry out Activity 3 in the **Participant Workbook** (page 7).



CASE EXAMPLES:

Identifying hazards/hazardous events



Consider these **real-world examples** that highlight the **importance of detailed and accurate system diagrams.....**

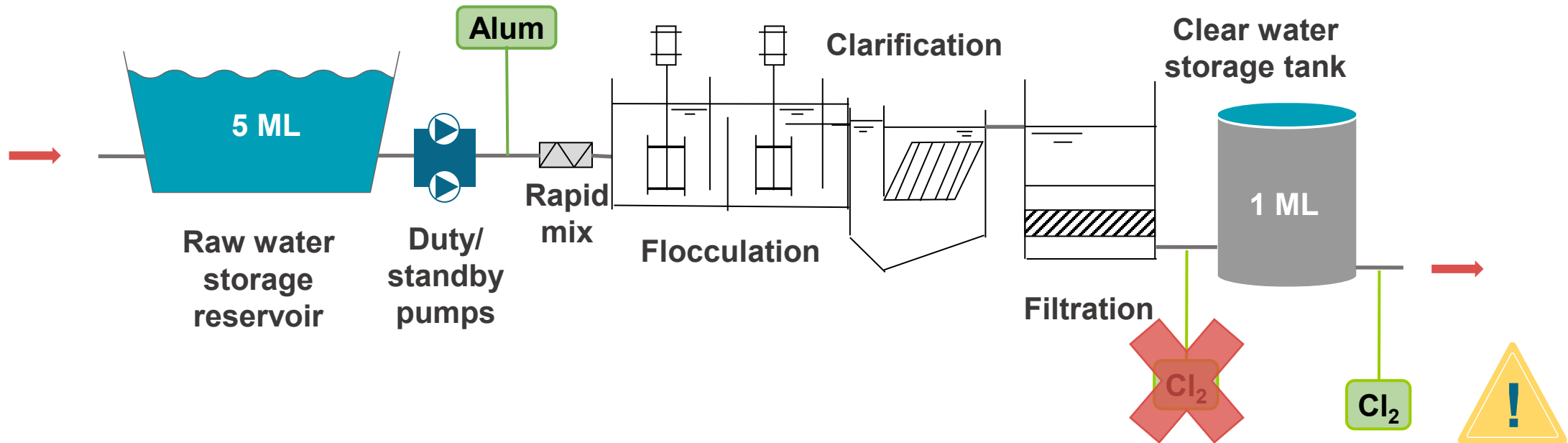


WHAT is the problem?



AUSTRALIA: Hazard detected by field verifying system diagram

1 ➡



Raw Water

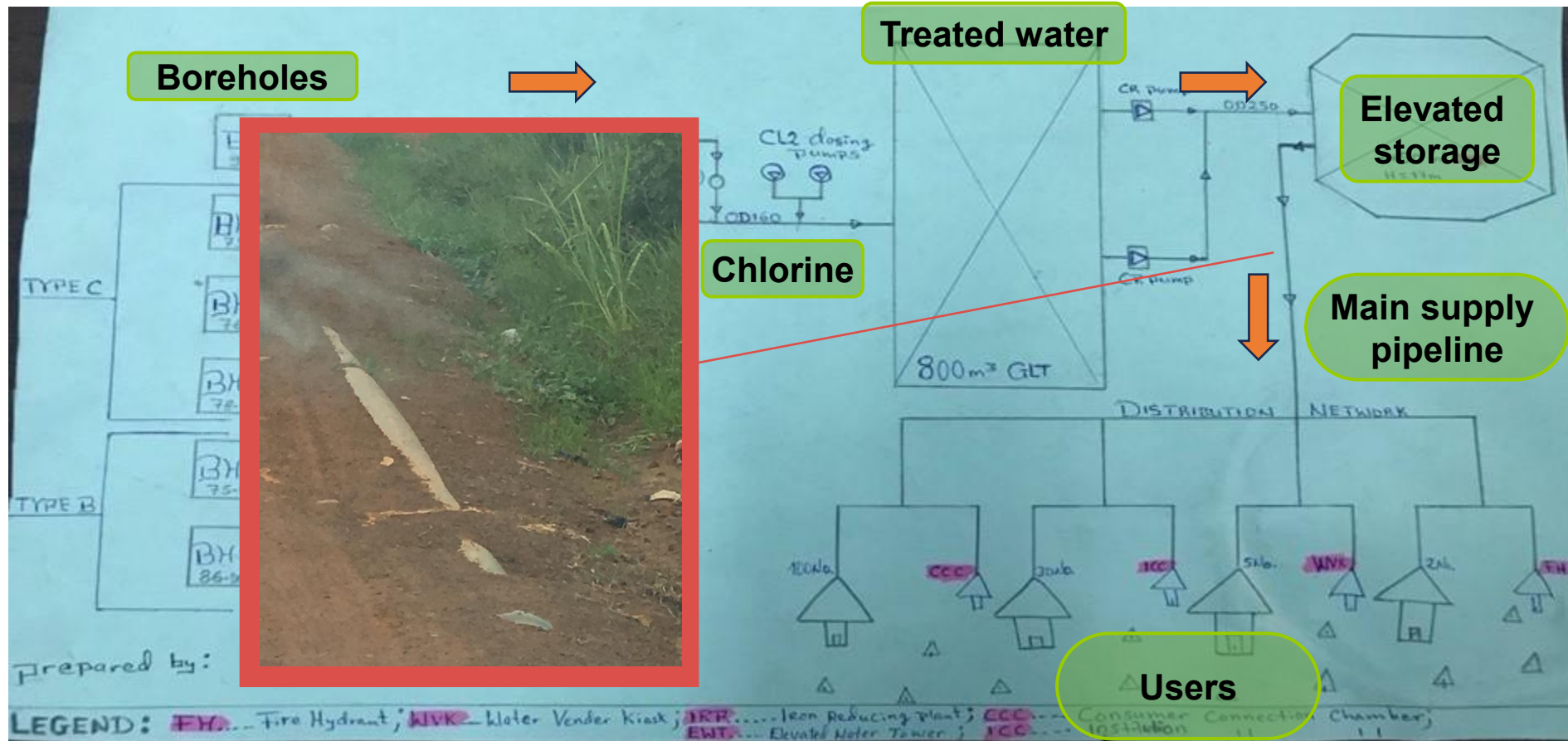
Treated Water

WHAT is the problem?



LIBERIA: Hazard detected by field inspection

1 ➡



Credit: WHO/RM McKeown

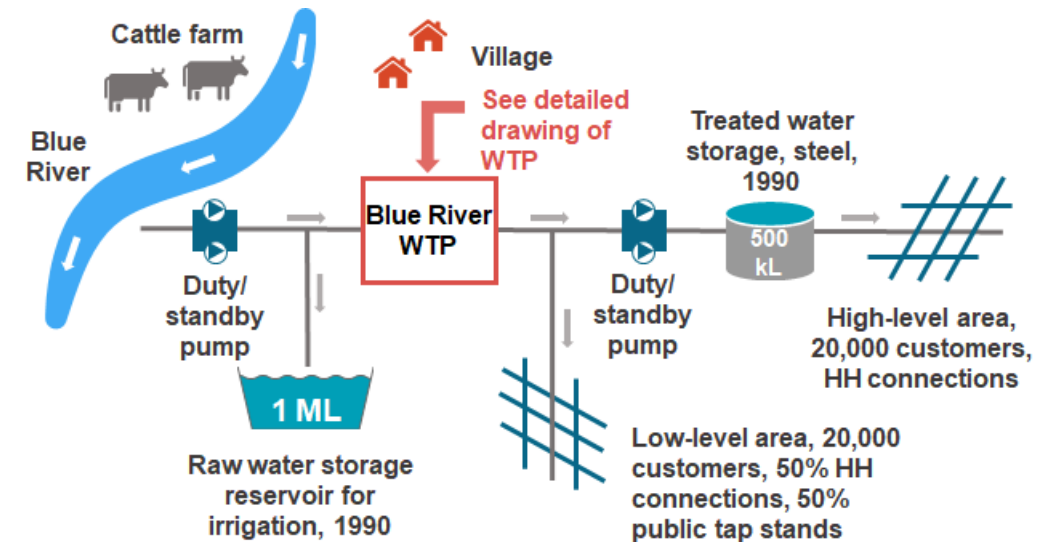
SYSTEM DIAGRAMS

support hazard identification



MORAL OF THE STORIES:

- Detailed and accurate system diagrams (Module 2) play an important role in hazard identification (Module 3)
- Desk-top identification of hazards/hazardous events alone is insufficient to detect all threats within a system → ***field visits are required!***



10 WSP modules

DAY 1	Mod 2: Describing the system
	Mod 3: Identifying hazards and hazardous events
DAY 2	Mod 4: Validating existing control measures and assessing risks
	Mod 5: Planning for improvement
	Mod 6: Monitoring control measures
	Mod 7: Verifying the effectiveness of water safety planning
DAY 4	Mod 8: Strengthening management procedures
	Mod 9: Strengthening WSP supporting programmes
	Mod 10: Reviewing and updating the WSP
	Mod 1: Assembling the WSP team

DISCUSSION and questions?





Credit: WHO/RM McKeown

That's all for today!